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Subject: ADBMS

Practical MST

Q1

Problem Statement 01

An online store maintains customers and their orders. Display customers whose **total purchase amount is greater than the total purchase amount of every other customer from the same city**.

Display **Customer Name, City, and Total Purchase Amount**. If two customers from the same city tie for the highest total, display both.

Constraint: Only a solution using JOINS and SUBQUERIES is allowed. CTEs and Window Functions are not allowed.

customer_id	customer_name	city
1	Aman	Delhi
2	Neha	Delhi
3	Raj	Mumbai
4	Priya	Mumbai
5	Karan	Pune

Customer

order_id	customer_id	amount
101	1	4000
102	1	3000
103	2	5000
104	3	6000
105	3	2000
106	4	8000
107	5	4500

Orders

Solution:

```
Lab_MST/postgres@PostgreSQL 18
Query History
1 CREATE TABLE Customer (
2   customer_id INT PRIMARY KEY,
3   customer_name VARCHAR(50),
4   city VARCHAR(50)
5 );
6
7 INSERT INTO Customer (customer_id, customer_name, city)
8 VALUES
9 (1, 'Aman', 'Delhi'),
10 (2, 'Neha', 'Delhi'),
11 (3, 'Raj', 'Mumbai'),
12 (4, 'Priya', 'Mumbai'),
13 (5, 'Karan', 'Pune');
14
15
Data Output Messages Notifications
INSERT 0 5
Query returned successfully in 198 msec.
```

Lab_MST/postgres@PostgreSQL 18

Query Query History

```

1 CREATE TABLE Orders (
2     order_id INT PRIMARY KEY,
3     customer_id INT REFERENCES Customer(customer_id),
4     amount INT
5 );
6
7 INSERT INTO Orders (order_id, customer_id, amount)
8 VALUES
9 (101, 1, 4000),
10 (102, 1, 3000),
11 (103, 2, 5000),
12 (104, 3, 6000),
13 (105, 3, 2000),
14 (106, 4, 8000),
15 (107, 5, 4500);

```

Data Output Messages Notifications

INSERT 0 7

Query returned successfully in 66 msec.

Lab_MST/postgres@PostgreSQL 18

Query Query History

```

1 SELECT
2     c.customer_name, c.city, sum(o.amount) AS total_purchase
3 FROM
4     Customer c JOIN Orders o
5     ON c.customer_id=o.customer_id
6 GROUP BY c.customer_name, c.city
7

```

Data Output Messages Notifications

Showing rows:

	customer_name character varying (50)	city character varying (50)	total_purchase bigint
1	Raj	Mumbai	8000
2	Karan	Pune	4500
3	Aman	Delhi	7000
4	Neha	Delhi	5000
5	Priya	Mumbai	8000

Q2.



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Problem Statement 02

An inventory system stores products in product_inventory. Create a trigger that automatically records a log whenever a product is **inserted or deleted**. Store Product ID, Product Name, action (PRODUCT ADDED/PRODUCT REMOVED), and current timestamp in product_audit.

input

product_id	product_name	price
101	Laptop	65000
102	Mobile	30000

Expected output

audit_id	product_id	product_name	action
1	103	Tablet	PRODUCT ADDED
2	101	Laptop	PRODUCT REMOVED

Solution:

```
Lab_MST/postgres@PostgreSQL 18
Query History
1 CREATE TABLE product_inventory (
2   product_id INT PRIMARY KEY,
3   product_name VARCHAR(50),
4   price INT
5 );
Data Output Messages Notifications
CREATE TABLE
Query returned successfully in 63 msec.
```

```
Lab_MST/postgres@PostgreSQL 18
Query History
1 CREATE TRIGGER on_insert
2 AFTER INSERT
3 ON product_inventory
4 FOR EACH ROW(
5   LOG(product_id, product_name, "Product Added");
6 )
7
8 CREATE TRIGGER on_delete
9 AFTER DELETE
10 ON product_inventory
11 FOR EACH ROW(
12   LOG(product_id, product_name, "Product Removed")
13 );
Data Output Messages Notifications
```